2 VISUALIZING DATA (PART 2)

2.1 TWO-WAY TABLES

Remember that one-way tables were made using **variable** data given for **individuals**. For example, **people and their heights**, or **ice cream flavors and the number of scoops sold**.

In this section we're transitioning to talk about **two-way tables**, which can be constructed from data that's dependent on two categorical variables.

Sometimes we talk about this kind of data in terms of **independent variables** and **dependent variables**.

In the case of data in one-way tables, we had one independent variable, called the individuals, and one or more dependent variables, called the variables. In the case of data in two-way tables, we have two independent categories on which the variables are dependent.

The difference between both types is easiest to visualize by comparing a data table for each one.

Suppose a small **coffee shop** opened in early **2019** in the Philippines. By the end of **2022**, it had been operating for **4** years, and its revenue steadily increased each year.

If we total the revenue earned for each month across all **4** years, we can summarize the data in a **one-way table**.

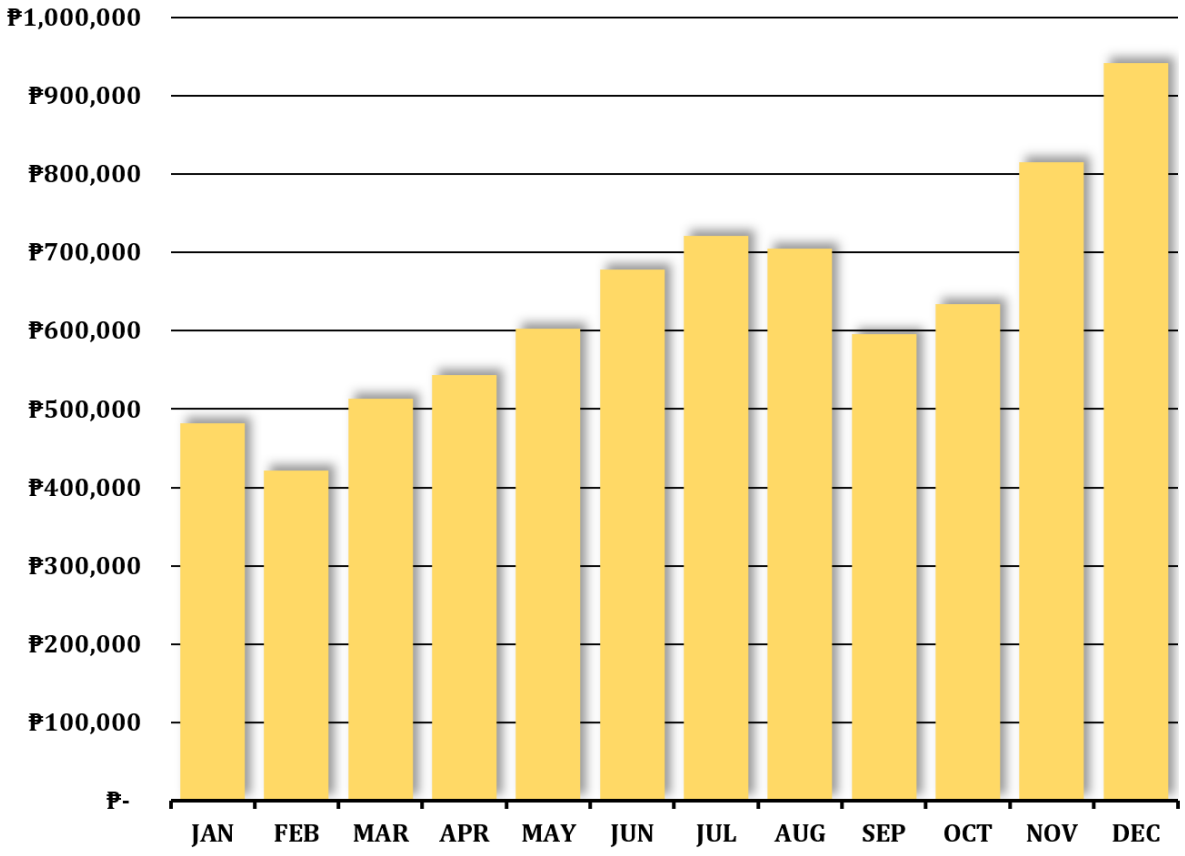
Month	Revenue
JAN	₱482,000
FEB	₱421,500
MAR	₱513,000
APR	₱544,000
MAY	₱602,500
JUN	₱678,000
JUL	₱721,000
AUG	₱704,500
SEP	₱596,000
OCT	₱634,000
NOV	₱815,500
DEC	₱942,000

In this one-way table, we have the **months** down the left side. Those are the **individuals (also called the independent variable)**. The **revenue** is the **variable (also called the dependent variable)**.

In this chart, the revenue the shop earned only depends on one thing: **the month**. We can only ask questions like:

- *How much has the artist earned in February?*
- *How much has the artist earned in September?*

Let's represent this as a bar graph.



This certainly gives us information about which months tend to be best for the coffeshop. **November** and **December** seem to be best. But since the data is summarized, adding the revenue for **January** for all **4 years** into one figure, and for **February**, for **March**, for **April**, etc., this one-way data doesn't give us any insight into how the business has grown over time.

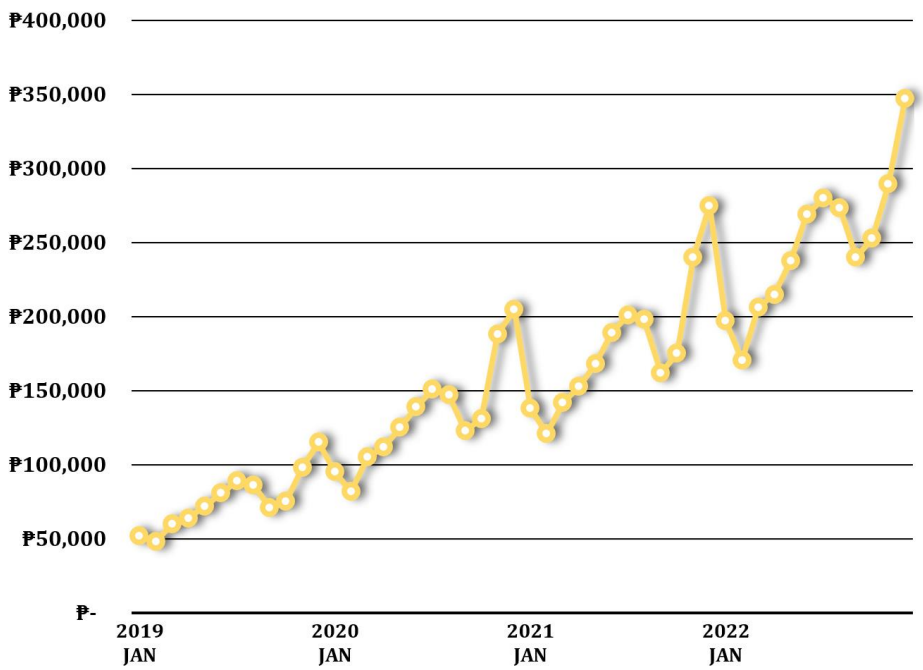
So to get more visibility, we could instead break apart the revenue information and show each year separately in a **two-way table**:

Month	2019	2020	2021	2022
JAN	₱52,000	₱95,000	₱138,000	₱197,000
FEB	₱48,000	₱82,000	₱121,000	₱170,500
MAR	₱60,000	₱105,000	₱142,000	₱206,000
APR	₱64,000	₱112,000	₱153,000	₱215,000
MAY	₱72,000	₱125,000	₱168,000	₱237,500
JUN	₱81,000	₱139,000	₱189,000	₱269,000
JUL	₱89,000	₱151,000	₱201,000	₱280,000
AUG	₱86,000	₱147,000	₱198,000	₱273,500
SEP	₱71,000	₱123,000	₱162,000	₱240,000
OCT	₱75,000	₱131,000	₱175,000	₱253,000
NOV	₱98,000	₱188,000	₱240,000	₱289,500
DEC	₱115,000	₱205,000	₱275,000	₱347,000

Notice how the data in the two-way table is now dependent on two independent things, not just one. In the one-way table, if someone asked us how much revenue the coffee shop earned, we'd reply *"For which month?"* But in the two-way table, if someone asked us how much revenue the coffee shop earned, we'd reply *"For which year and for which month?"*

In the two-way table, each value depends on two things: **the month and the year**. Whereas in the one-way table, each value depended on only on thing: **the month**.

We could graph the data in the two-way table in lots of different ways, but let's do it as a **line graph** that starts in **January, 2019** and ends in **December, 2022**.



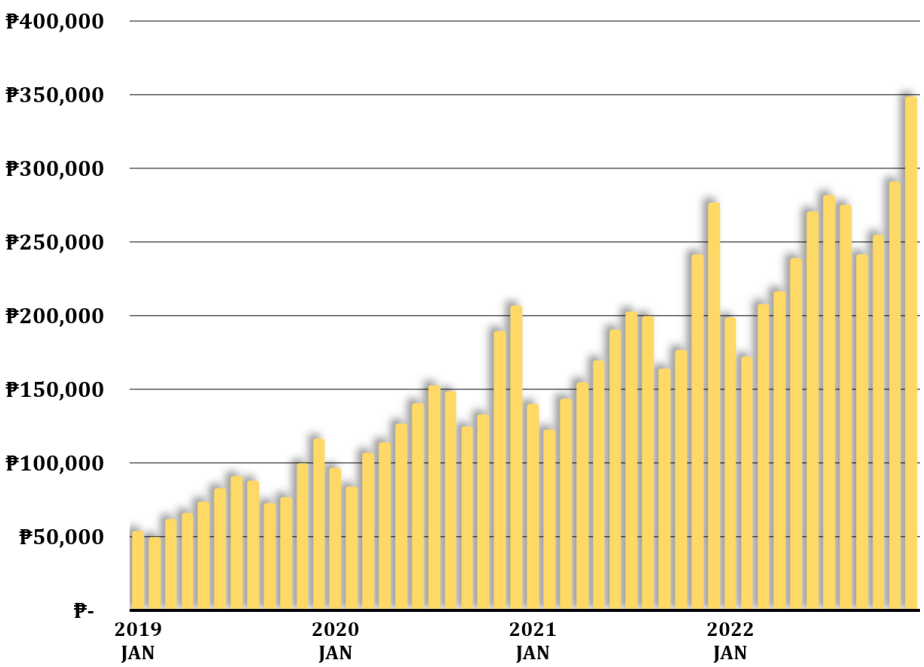
By giving the coffee shop revenue data in a two-way table, we get so much more insight into the business

2.1.1 SHOWING TOTALS

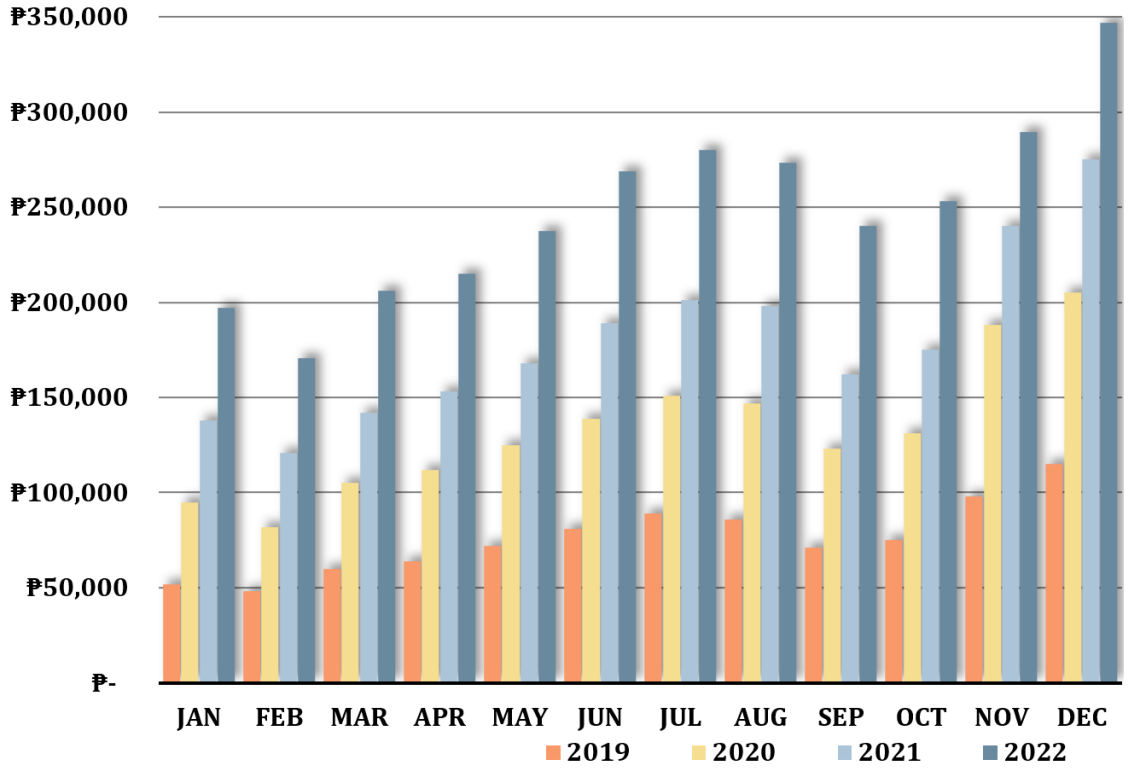
In two-way tables, we can also summarize the data by row and by column, plus give a total for the entire table in the lower right-hand corner.

Month	2019	2020	2021	2022	Total
JAN	₱52,000	₱95,000	₱138,000	₱197,000	**₱482,000**
FEB	₱48,000	₱82,000	₱121,000	₱170,500	**₱421,500**
MAR	₱60,000	₱105,000	₱142,000	₱206,000	**₱513,000**
APR	₱64,000	₱112,000	₱153,000	₱215,000	**₱544,000**
MAY	₱72,000	₱125,000	₱168,000	₱237,500	**₱602,500**
JUN	₱81,000	₱139,000	₱189,000	₱269,000	**₱678,000**
JUL	₱89,000	₱151,000	₱201,000	₱280,000	**₱721,000**
AUG	₱86,000	₱147,000	₱198,000	₱273,500	**₱704,500**
SEP	₱71,000	₱123,000	₱162,000	₱240,000	**₱596,000**
OCT	₱75,000	₱131,000	₱175,000	₱253,000	**₱634,000**
NOV	₱98,000	₱188,000	₱240,000	₱289,500	**₱815,500**
DEC	₱115,000	₱205,000	₱275,000	₱347,000	**₱942,000**
Total	**₱911,000**	**₱1,603,000**	**₱2,162,000**	**₱2,978,000**	**₱7,654,000**

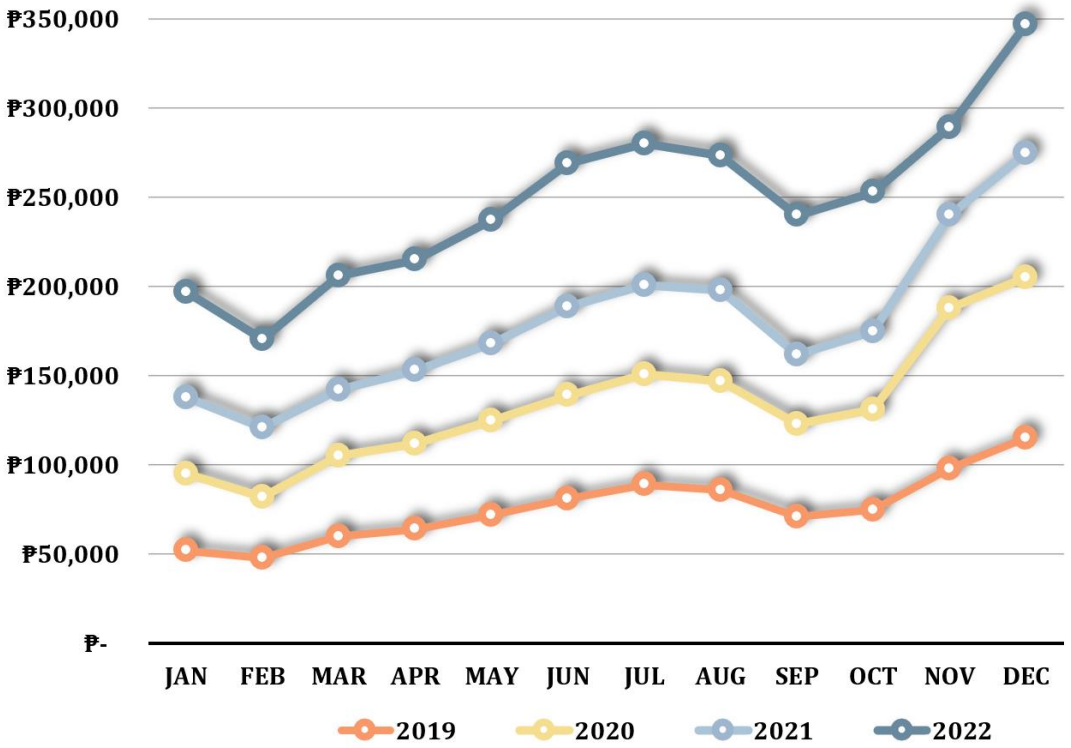
From the two-way table, we can now see the revenue totals for each month down the right side of the table (these match the totals in the one-way table), but we can also see the revenue totals for each year along the bottom. We can still see which month is best for the business (**December**), but we can also see which year was best (**2022**). We did a line graph to represent all of this data over time, but we could have done a **bar graph**, instead.



We could also create a **comparison bar graph**, illustrating the revenue totals with side-by-side color coded bars for each year,

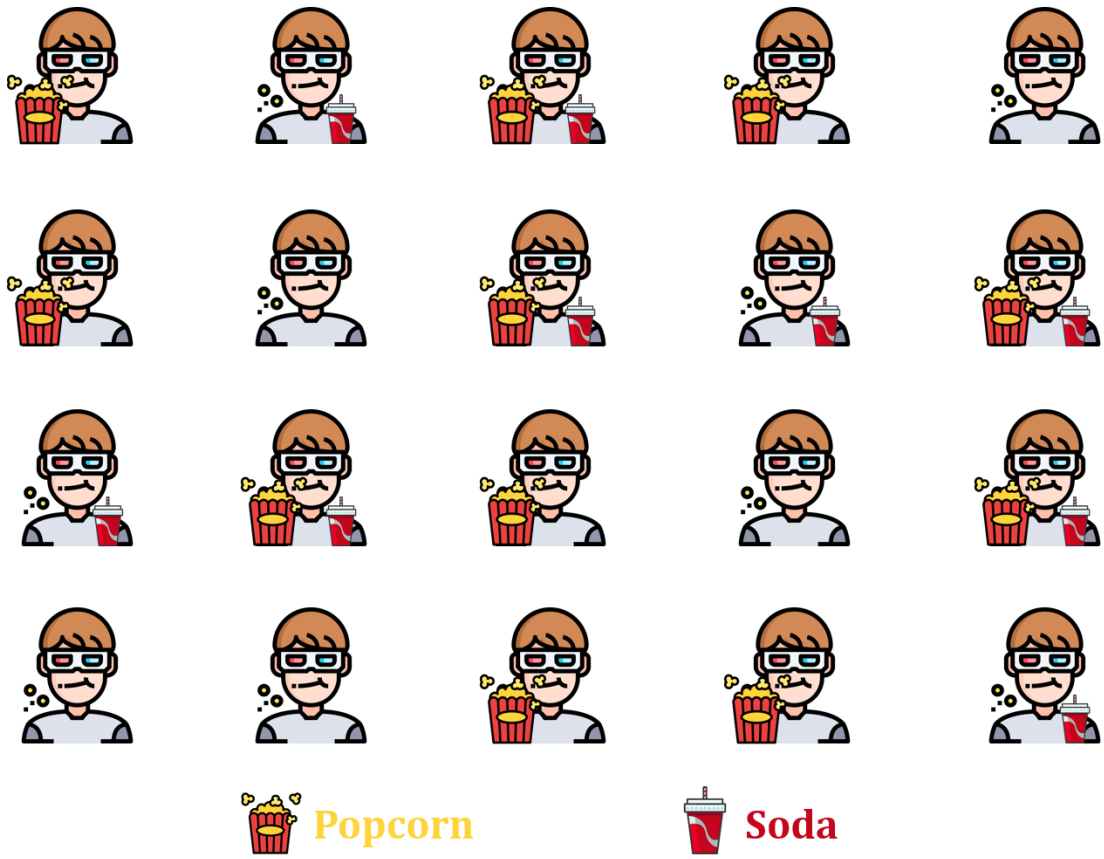


or a **comparison line graph**, depicting color-coded line graphs for each year.



2.2 VENN DIAGRAMS

We've seen how we can express data in a **two-way table** and then translate that data into bar and line graphs. However, we can also represent data from a two-way table using a different visualization called a **Venn diagram**. Let's say it is the premiere night of a movie, and the stalls are selling popcorn and soda. We want to categorize the goods sold by creating a chart that shows how many people bought each type of snack. Here is a picture representing all of our moviegoers:

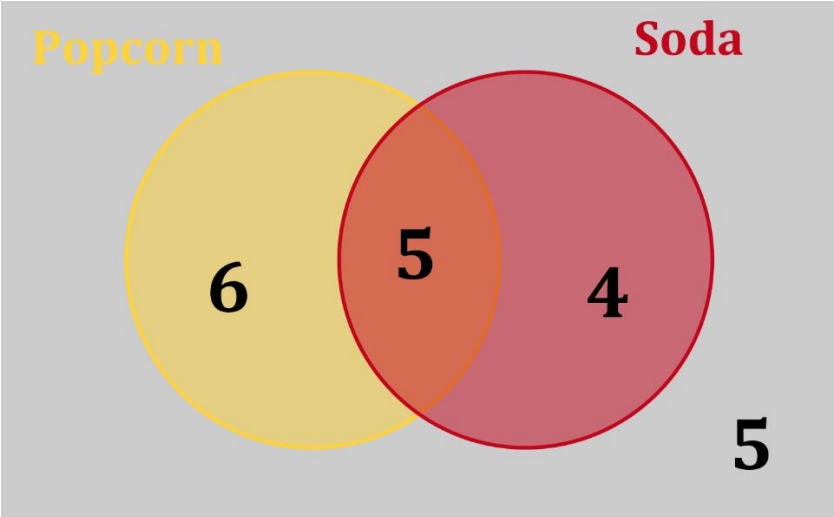


As we can see, some of or moviegoers have bought popcorn and soda, and some don't have either popcorn or soda.

Let's apply what we already know and represent this data in a **two-way table**. We'll go ahead and include totals as well.

	Soda	No Soda	**Total**
Popcorn	5	6	**11**
No Popcorn	4	5	**9**
Total	**9**	**11**	**20**

A **Venn diagram** representing this data might look like this:



Let's walk through this. The **red circle** represents the **moviegoers that bought soda** the **yellow circle** represents the **moviegoers with popcorn**.

- Because we put a **4** inside the **red circle** but outside the **yellow circle**, it means there are **4** moviegoers that has **soda** but **no popcorn**.
- Because we put a **6** inside the **yellow circle** but outside the **red circle**, it means there are **6** moviegoers that have **popcorn** but **no soda**.
- Because we put a **5 inside both circles in the middle**, it means there are **5** moviegoers that have both **popcorn** and **soda**.
- Because we put a **5 outside both circles in the corner**, it means there are **5** moviegoers that didn't have buy anything.

From the Venn diagram, we can also see that there are

- **6 + 5 = 11** total moviegoers with popcorn,
- **4 + 5 = 9** total moviegoers with soda,
- **6 + 4 + 5 = 15** total of moviegoers that have at least bought a popcorn or soda or both, and
- **6 + 5 + 4 + 5 = 20** moviegoers in total

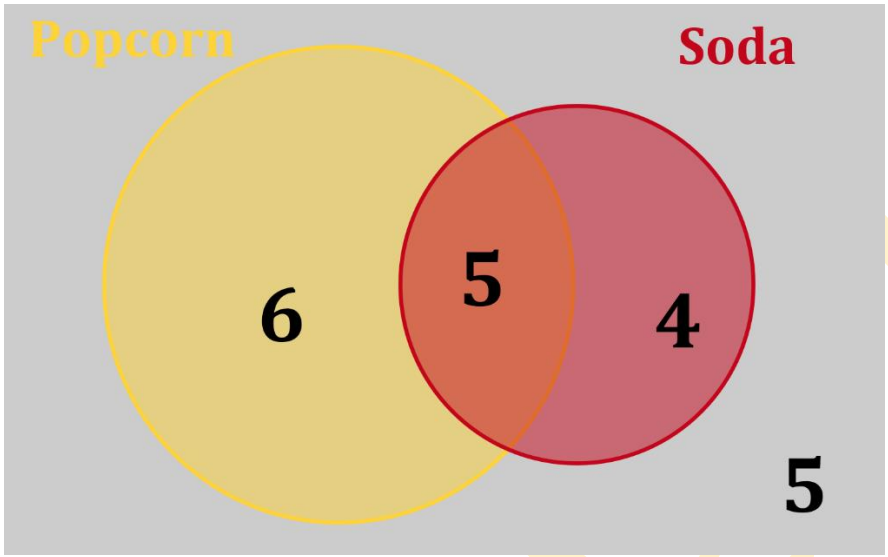
Notice that all of these numbers match the data in the two-way table.

So when we construct a Venn diagram, we put the number of items that only meet one condition in one circle, we put the number of items that only meet the other condition in the other circle, we put the number of items that meet both conditions in the middle where the circles overlap, and we put the number of items that meet neither condition outside of all the circles but inside the box.

2.2.1 VARIATIONS OF VENN DIAGRAMS

We can also construct Venn diagrams with circles of different sizes, where the size of each circle is proportional to the number of items it represents.

So if we take the Venn diagram we made earlier for the moviegoers, we could adjust the circles so that the ratio of the popcorn circle to the soda circle is **11 : 9**

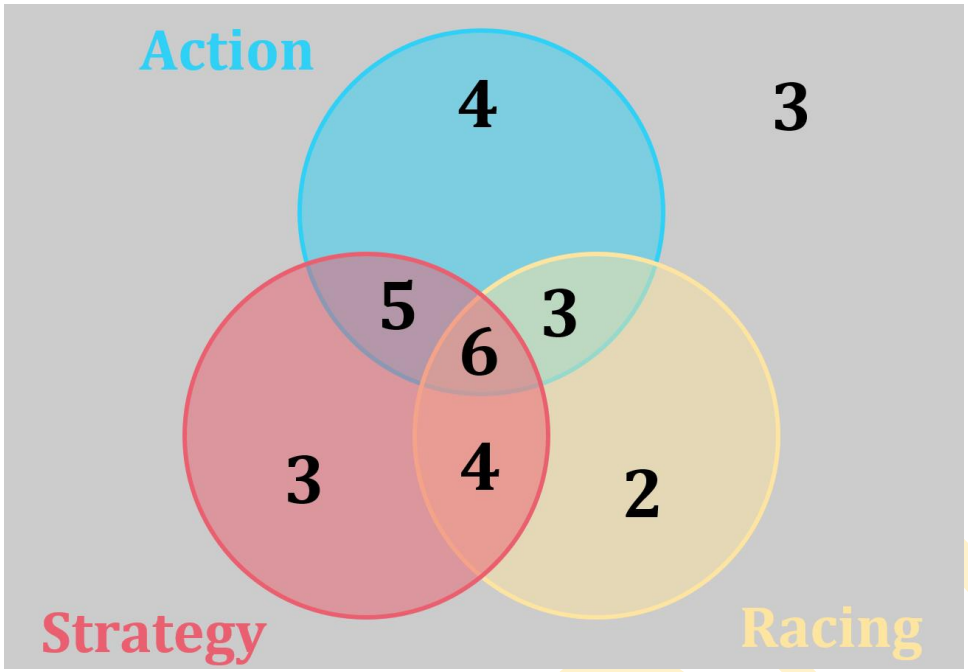


We can also create Venn diagrams with more than just two circles. For instance, suppose we asked students about their favorite game genre, and collected the following data.

- **4** students like **Action** only
- **3** students like **Strategy** only
- **2** students like **Racing** only
- **5** students like **Action and Strategy** only
- **3** students like **Action and Racing** only
- **4** students like **Strategy and Racing** only
- **6** students like **all three**
- **3** students like **none**

Even though there are three kinds of game genre in this survey, **Action, Strategy, and Racing**, we can still construct a Venn diagram that models student preferences.

Let's use a **blue circle** for students who like **Action**, a **red circle** for students who like **Strategy**, and **yellow circle** for students who like **Racing**.



From the diagram, we can see that,

- The **4** inside the **blue circle**, but outside the **red circle** and **yellow circle**, represents students who **like Action only**.
- The **3** inside the **red circle**, but outside the **blue circle** and **yellow circle**, represents students who **like Strategy only**.
- The **2** inside the **yellow circle**, but outside the **blue circle** and **red circle**, represents students who **like Racing only**.
- The **5** located in the **overlap of the blue and red circle**, but outside the **yellow circle**, represents students who **like Action and Strategy, but not Racing**.
- The **3** in the **overlap of the blue and yellow circle**, but outside the **red circle**, represents students who **like Action and Racing, but not Strategy**.
- The **4** in the **overlap of the red and yellow circle**, but outside the **blue circle**, represents students who **like Strategy and Racing, but not Action**.
- The **6** in the center where **all three circles overlap** represents students who **like Action, Strategy, and Racing**.
- The **3** placed **outside all three circles** represents students who **do not like any of the three game genres**.

2.3 FREQUENCY TABLES AND DOT PLOTS

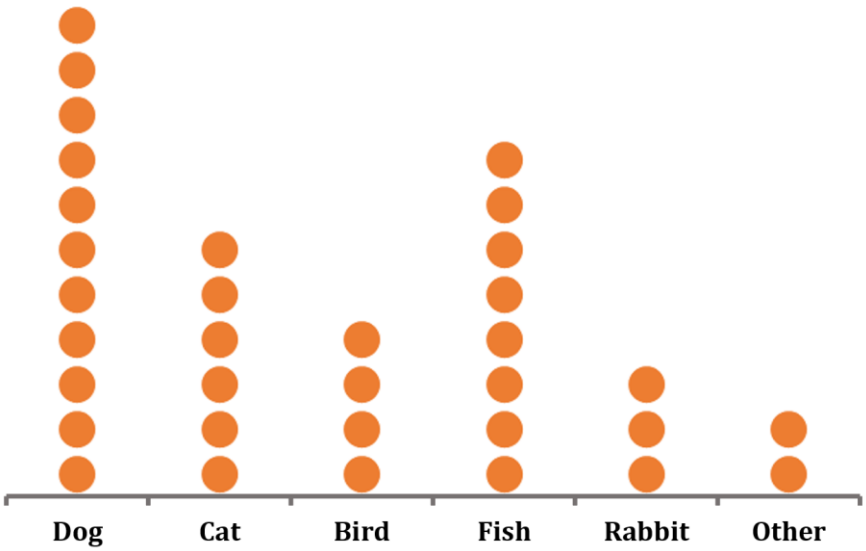
Think of a frequency table as a table that shows how often or how rarely something happens. A **dot plot** can also be used to represent the frequency of small data sets.

If I want to categorize the **pets in a neighborhood by type**, I could organize them in a frequency table.

Type	Count
Dog	11
Cat	6
Bird	4
Fish	8
Rabbit	3
Other	2

To create this table, I counted how many pets belonged to each type and recorded the totals.

I could also display this information using a dot plot.



To construct the dot plot, we simply convert the counts from the frequency table into dots. Because there are **11 dogs** listed in the table, we would place **11 dots** above **Dog** in the dot plot.

Dot plots are similar to bar graphs because we can quickly identify which category appears most often by looking at the height of the dots, just like comparing bar heights in a bar graph.

We can also convert between a raw data list, a frequency table, and a dot plot.

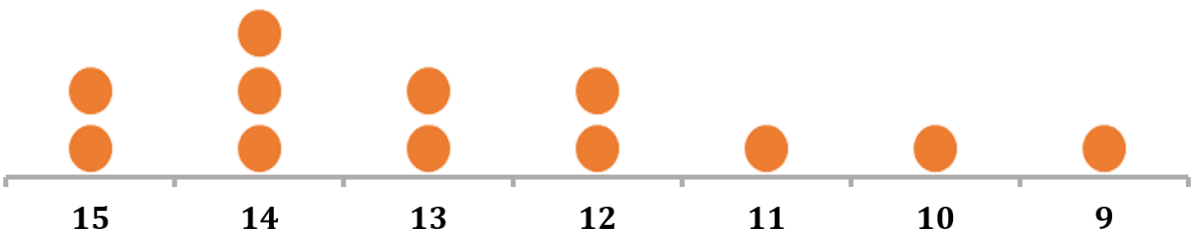
For example, **12 students** participated in a short spelling quiz where scores ranged from **0 to 15**. Their scores were:

14, 12, 15, 10, 14, 13, 15, 9, 12, 14, 11, 13

A frequency table can be created to show how many times each score occurred.

Quiz Score	Count
15	2
14	3
13	2
12	2
11	1
10	1
9	1
8	0
7	0
6	0
5	0
4	0
3	0
2	0
1	0
0	0

The data from this frequency table can then be converted into a dot plot by placing the corresponding number of dots above each score.



2.4 RELATIVE FREQUENCY TABLES

We're familiar now with displaying data in a two-way table. For example, we might categorize the books at *Maria's bookstore* by type and by format.

	Paperback	Hardcover
Fiction	18	12
Nonfiction	6	4

But sometimes it's helpful to express the data in a two-way table as percentages. For example, of all the paperback books in *Maria's bookstore*,

- **What percentage of them are *fiction*?**,
- and **what percentage of them are *nonfiction***?

If we want to express these percentages, then we just need to change the two-way table into what's called a **relative frequency table**, which is a table that shows percentages instead of actual counts.

So to answer our question from earlier, we know from our table that there are **18 + 6 = 24** total **paperback books**, which means that

- **18/24 = 0.75 = 75%** of the **paperback books** are **fiction**.
- **6/24 = 0.25 = 25%** of the **paperback books** are **nonfiction**.

We could do the same calculations for the **hardcover books**:

- **Fiction**: **12/(12 + 4) = 12/16 = 0.75 = 75%**
- **Nonfiction**: **4/(12 + 4) = 4/16 = 0.25 = 25%**

So we could change the table from one that just shows a count of books, into a relative frequency table that shows the percentage of fiction and nonfiction based on format.

	Paperback	Hardcover
Fiction	0.75	0.75
Nonfiction	0.25	0.25
Column Total	**1.0**	**1.00**

Now we can see that out of all the **paperback books**, **75%** of them are **fiction** and **25%** of them are **nonfiction**. And out of all the **hardcover books**, **75%** are **fiction** and **25%** are **nonfiction**.

We could also create a relative frequency table from a one-way table. Using the table from a previous section about the number of students enrolled in different school clubs,

Club	Count
Drama	12
Science	9
Music	5
Sports	14
Column Total	**40**

We could turn this table into a relative frequency table.

Club	Count
Drama	0.30
Science	0.23
Music	0.13
Sports	0.35
Column Total	**1.00**

2.4.1 ROWS VS. COLUMNS

Notice that because we found the percent of the total within each column, that we were able to add a **Column total** to the bottom of the book table. Each column sums to **1.00**.

- **0.75 + 0.25 = 1.00**
- **0.75 + 0.25 = 1.00**

That's because in each cell, we found the percentage of the total for the column, and so each column sums up to **100%**. Therefore, the table we made is actually called a **column-relative frequency**, where the columns sum to **1.00** but the rows do not.

But we could have just as easily created a row-relative frequency by converting the rows into percentages instead of the columns into percentages.

We'll start again with the original table.

	Paperback	Hardcover
Fiction	18	12
Nonfiction	6	4

This time, we want to find the percentage of each format based on the kind of book. In other words, what percentage of **fiction books** are **paperback** or **hardcover**, and what percentage of **nonfiction** books are **paperback** or **hardcover**? Let's do the calculations for **fiction**:

- **Paperback**: **18/(18 + 12) = 18/30 = 0.60 = 60%**
- **Hardcover**: **12/(18 + 12) = 12/30 = 0.40 = 40%**

Now we'll do the calculations for **nonfiction**:

- **Paperback**: **6/(6 + 4) = 6/10 = 0.60 = 60%**
- **Hardcover**: **4/(6 + 4) = 4/10 = 0.40 = 40%**

Now we can turn the table into a **row-relative frequency table**.

	Paperback	Hardcover	Row Total
Fiction	0.60	0.60	**1.00**
Nonfiction	0.40	0.40	**1.00**

Out of all the **fiction books**, **60%** of them are **paperback** and **40%** are **hardcover**. And out of all the **nonfiction books**, about **60%** are **paperback** and about **40%** are **hardcover**.

2.4.2 TOTAL RELATIVE FREQUENCY

We can also calculate the **total-relative frequency** for a two-way table by dividing everything by the **grand total**.

	Paperback	Hardcover	Total
Fiction	18	12	**30**
Nonfiction	6	4	**10**
Total	**24**	**16**	**40**

Since there are **40** total books, we divide each value in the body of the table and each value in the total column and row by **40**. Then we can express the values as **decimals**

	Paperback	Hardcover	Total
Fiction	0.45	0.30	**0.75**
Nonfiction	0.15	0.10	**0.25**
Total	**0.60**	**0.40**	**1.00**

or as **percentages**.

	Paperback	Hardcover	Total
Fiction	45%	30%	**75%**
Nonfiction	15%	10%	**25%**
Total	**60%**	**40%**	**100%**

2.5 JOINT DISTRIBUTIONS

A **joint distribution** is a table of percentages similar to a relative frequency table. The difference is that, in a joint distribution, we show the distribution of one set of data against the distribution of another set of data.

Let's say we study a group of **100** students, measuring the **average number of hours** each student spent studying per week over the course of the term, and we also gathered data about the **total number of quiz points** earned by each student over that same period of time.

		Quiz Points			
		0-10	**10-20**	**20-30**	**30+**
Hours Studying	**0-3**	3%	2%	1%	1%
	3-6	7%	5%	4%	1%
	6-9	2%	14%	9%	5%
	9-12	4%	4%	11%	8%
	12+	4%	5%	5%	5%

From this table, we can see that **3%** of the group, which would be **3 out of the 100 students** studied, spent between **0-3 hours** per week studying and earned between **0-10 quiz points**. More students in the study (**14% or 14 out of the 100**) studied between **6-9 hours** per week on average and earned between **10-20 quiz points** in total.

This is an example of a joint distribution because we're comparing the distribution of two distributions:

- the distribution of average weekly hours spent studying,
- and the distribution of total quiz points earned over the course of the term.

When we read a joint distribution table, we'll oftentimes look at **marginal and conditional distributions** within the table.

2.5.1 MARGINAL DISTRIBUTION

We could total up the data in each row and each column, and add those totals to the table:

		Quiz Points				
		0-10	10-20	20-30	30+	Total
Hours Studying	0-3	3%	2%	1%	1%	7%
	3-6	7%	5%	4%	1%	17%
	6-9	2%	14%	9%	5%	30%
	9-12	4%	4%	11%	8%	27%
	12+	4%	5%	5%	5%	19%
	Total	20%	30%	30%	20%	100%

Think of a **marginal distribution** as the **Total column** or the **Total row** in this joint distribution. It's like only having one of the distributions, not both. So if we only had the distribution of quiz points, we'd have just the totals along the bottom of the table.

	Quiz Points				
	0-10	10-20	20-30	30+	Total
Total	20%	30%	30%	20%	100%

This table gives us the **marginal distribution for quiz points**; it tells us the percentage of students who earned **0-10 points**, **10-20 points**, etc. But it doesn't give us any information about how much they studied.

We could also look at only the distribution for hours spent studying, taking the totals along the right side of the original joint distribution.

		Total
Hours Studying	0-3	7%
	3-6	17%
	6-9	30%
	9-12	27%
	12+	19%
	Total	100%

This is the marginal distribution for hours of studying, and we can see the percentage of students who studied **0-3 hours**, **3-6 hours**, etc. But this table doesn't give us any information about how many quiz points were earned based on how much time was spent studying.

2.5.2 CONDITIONAL DISTRIBUTION

Think of a **conditional distribution** as the distribution of one variable, given a particular value of the other variable.

For example, we could look at the **conditional distribution of quiz points**, given a particular amount of hours spent studying:

		Quiz Points				
		0-10	**10-20**	**20-30**	**30+**	**Total**
Hours Studying	**0-3**	42.86%	28.56%	14.29%	14.29%	**100%**
	3-6	41.18%	29.41%	23.53%	5.88%	**100%**
	6-9	6.67%	46.67%	30%	16.66%	**100%**
	9-12	14.81%	14.81%	40.75%	29.63%	**100%**
	12+	21.04%	26.32%	26.32%	26.32%	**100%**

Looking at the top row of this conditional distribution, we can say this:

Given that students studied **0-3 hours**,

- **43% of them earned 0-10 quiz points**
- **29% of them earned 10-20 quiz points**
- **14% of them earned 20-30 quiz points**
- **14% of them earned 30+ quiz points**

But this distribution is conditional on **0-3 hours** spent studying. Notice that the distributions in the table are conditional upon the value of hours spent studying in each row, so we see the row totals at **100%**.

Or we could look at the **conditional distribution of hours studying**, given a particular amount of quiz points:

		Quiz Points			
		0-10	**10-20**	**20-30**	**30+**
Hours Studying	**0-3**	15%	6.66%	3.33%	5%
	3-6	35%	16.67%	13.33%	5%
	6-9	10%	46.67%	30%	25%
	9-12	20%	13.33%	36.67%	40%
	12+	20%	16.67%	16.67%	25%
	Total	**100%**	**100%**	**100%**	**100%**

Looking at the first column of this conditional distribution, we can say this:

Given that students earned 0-10 quiz points,

- **17% studied 0-3 hours**
- **39% studied 3-6 hours**
- **11% studied 6-9 hours**
- **22% studied 9-12 hours**
- **11% studied 12+ hours**

But this distribution is conditional on **0-10 quiz points** earned. Notice that the distributions in the table are conditional upon the value of quiz points in each column, so we see the column totals at **100%**.

2.6 HISTOGRAM AND STEM-AND-LEAF PLOTS

2.6.1 HISTOGRAM

A **histogram**, also called a **frequency histogram**, is just like a bar graph, except that we collect the data into buckets or bins, and then sketch a bar for each bucket. Each bucket needs to be the same size, or width, so that they're capable of holding the same amount of data.

Unlike bar charts, histograms **have no gaps between the bars** (although some bars might be absent, which means there's no frequency in that *"bucket"*). A histogram represents a **continuous dataset**, which is why there are no gaps between the buckets.

One reason we might want to use a histogram instead of a bar graph is because **we have too many data points to plot individually**.

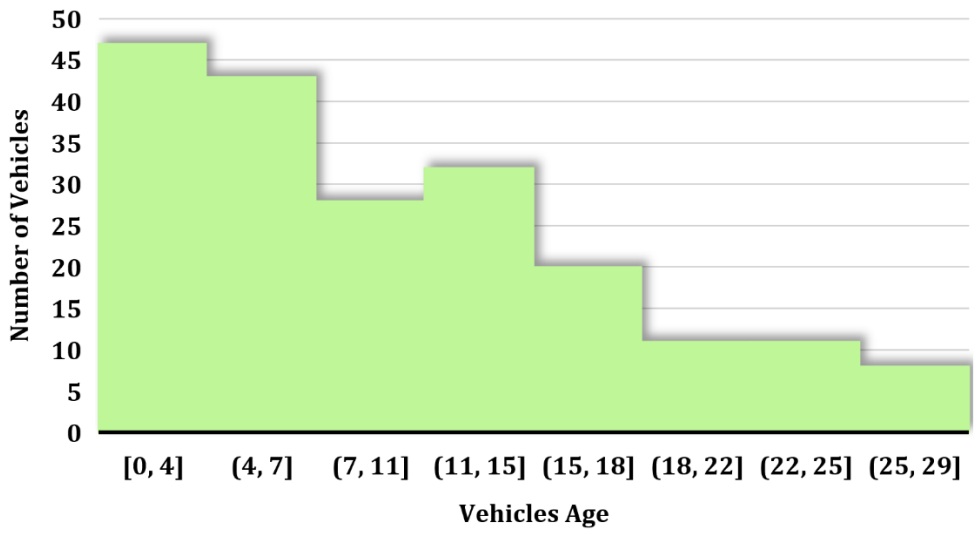
For example, maybe we want to use city records to make a graph of *the number of vehicles of each age in the entire city of Manila*. In a typical bar graph, we would have to show a bar for cars younger than **1 year old**, another for **1-year-old cars**, for **2-year-old cars**, **3**, **4**, **5**, all the way up to **30** or maybe even older. In other words, our bar graph might have **30** bars or more.

A **histogram** is the perfect solution to an overly-complicated bar graph. To create a histogram for the same information, we might group together

- **0-4 year-old cars**,
- **5-9 year-old cars**,
- **10-14 year-old cars**,
- etc.

Notice that each of these buckets is the same size or length. That's important to remember when making a histogram. Putting vehicles of similar age together in those groups would allow us to create a histogram with around **8 bars**, instead of a bar graph with around **30 bars**.

The histogram might look like this:



The way the data is spread out in the histogram is called the **distribution**.

As a very general rule,

- **qualitative data is usually better in a bar graph**, and
- **quantitative data is usually better in a histogram**.

2.6.2 STEM-AND-LEAF PLOT

A **stem-and-leaf plot** (also called a **stem plot**) is just another way to summarize data. It's similar to a histogram, because both types of charts group together data points and are good ways to show how many data points fall into a certain category or range.

For stem-and-leaf plots, we group data together by the first digit(s) in each number. In other words, let's say we have the **test scores of students in a mathematics class**:

72, 73, 74, 75, 75, 76, 76, 77, 78, 78, 78, 79, 80, 81, 82, 83, 85, 88

We could create a stem plot of the scores.

7	2, 3, 4, 5, 5, 6, 6, 7, 8, 8, 8, 9
8	0, 1, 2, 3, 5, 8

Key: 7|2 = 72

Let's use this particular plot to talk about what a stem plot shows. First, the *"stems"* are the numbers on the left, in this case the **7** and the **8**. The *"leaves"* are all the other numbers on the right.

Each leaf represents one data point, in this case one test score. So if we want to know how many data points are in the set, we could count the number of leaves on the right side. In this plot, there are **18 leaves**, which means we collected **18 test scores**.

Notice that we also put **7|2 = 72** below the stem-and-leaf plot. This is a key, or legend, that tells us that we intended for the stem to represent the tens place, and for the leaf to represent the ones place. If the key had said **7|2 = 702**, that would have meant that each stem represented the hundreds place (**7 would indicate 700**), and the leaf would represent the ones place (**2 would indicate 2**).

Each leaf needs to be attached to the stem from the same row in order to give us each data point. In other words, the stem isn't a data point on its own, and neither is the leaf. They only make a data point when we put them together. So if we take the first stem, **7 tens**, and the first leaf, **2 ones**, we put them together to get **72**, and that's one test score. If we put the second leaf, **3 ones**, with the stem, we get **73**, which is another test score in our data set. We could do this up to the last leaf on that line, to get test scores from the first line of our plot of

$$72, 73, 74, 75, 75, 76, 76, 77, 78, 78, 78, 79$$

We could do the same with the second line to see that we also have scores of

$$80, 81, 82, 83, 85, 88$$

2.6.3 NO DIGIT, OR MORE THAN ONE DIGIT IN THE STEM?

In a stem-and-leaf plot, the leaf will always only have one digit, and the stem will take the rest of the digits. So for example, if another student scored **104** and we wanted to add that score into our plot, we'd make the **4** the leaf, and the **10** would be the stem.

7	2, 3, 4, 5, 5, 6, 6, 7, 8, 8, 8, 9
8	0, 1, 2, 3, 5, 8
10	4

Even single-digit numbers can be included in a stem plot. If we wanted to add data points of **6**, **8**, and **9** to our plot, we can either leave the stem blank,

	6, 8, 9
7	2, 3, 4, 5, 5, 6, 6, 7, 8, 8, 8, 9
8	0, 1, 2, 3, 5, 8

or put a **0** in the stem spot.

0	6, 8, 9
7	2, 3, 4, 5, 5, 6, 6, 7, 8, 8, 8, 9
8	0, 1, 2, 3, 5, 8

2.7 BUILDING HISTOGRAMS FROM DATASETS

Now that we understand the basics of histograms, let's dive into the details about how we can actually build a histogram if we're only given the data set. Let's start by defining some important terms related to histograms.

The first one is **class interval**, or **class**, or **bin**, as we mentioned in the previous lesson. We always divide our data set into class intervals with equal class width. The **class width** is usually the difference between either the upper limits of two consecutive classes or between the lower limits of two consecutive classes.

For example, if the first class in our histogram is **20–24** and the second class is **25–29**, then the class width is given by the difference between the upper limits, **29 − 24 = 5**, or by the difference between the lower limits, **25 − 20 = 5**.

A **class midpoint** is the value that's at the center of a particular class. The class midpoint is half the sum of the lower and upper limits of the class. For example, for the class **40–44**, the class midpoint is

$$\frac{40 + 44}{2} = 42$$

If at all possible, it's nice to choose a class width that's odd (like a width of **5**, **11**, **17**, *etc.*) because it'll make the class midpoint a whole number, instead of a decimal number.

Let's look at the steps that we need to use to turn raw data into a histogram.

Put the data set in **ascending order**, then find the range as the difference between the largest and smallest values.

Determine the number of bins, or classes, that we want to have in our histogram. As a rule of thumb, it's best to use $5-6$ classes for most of the data we'll work with during our statistical studies. However, we might want to use up to 20 classes when we deal with larger data sets. It all depends on how large our data set is and the number of classes that would best represent the data.

Divide the range by the number of classes, then round up the result to get the class width.

Build a table, putting each class in a separate row.

Find the frequency for each class by counting the data points that fall into each one. When adjacent bins are for intervals like $10-14$ and $14-18$, the data point 14 should be included in the second bin. In other words, the interval $10-14$ actually contains data from $10-13.999\ldots$, and the interval $14-18$ contains data from $14-17.999\ldots$

Graph the histogram by placing the classes along the horizontal axis and their frequencies along the vertical axis, such that the height of each bar is the frequency of each class.

Let's work through an example so that we can see these steps in detail.

Example

A total of 20 employees recorded the number of minutes they spent exercising last week. Construct a histogram for the number of exercise minutes spent by the group.

$$12, 25, 18, 30, 22, 15, 27, 10, 20, 17, 16, 24, 28, 35, 14, 19, 23, 32, 21, 26$$

First, put the values in ascending order. This makes it easier to see the $maximum$ and $minimum$ values of the data set.

$$10, 12, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 30, 32, 35$$

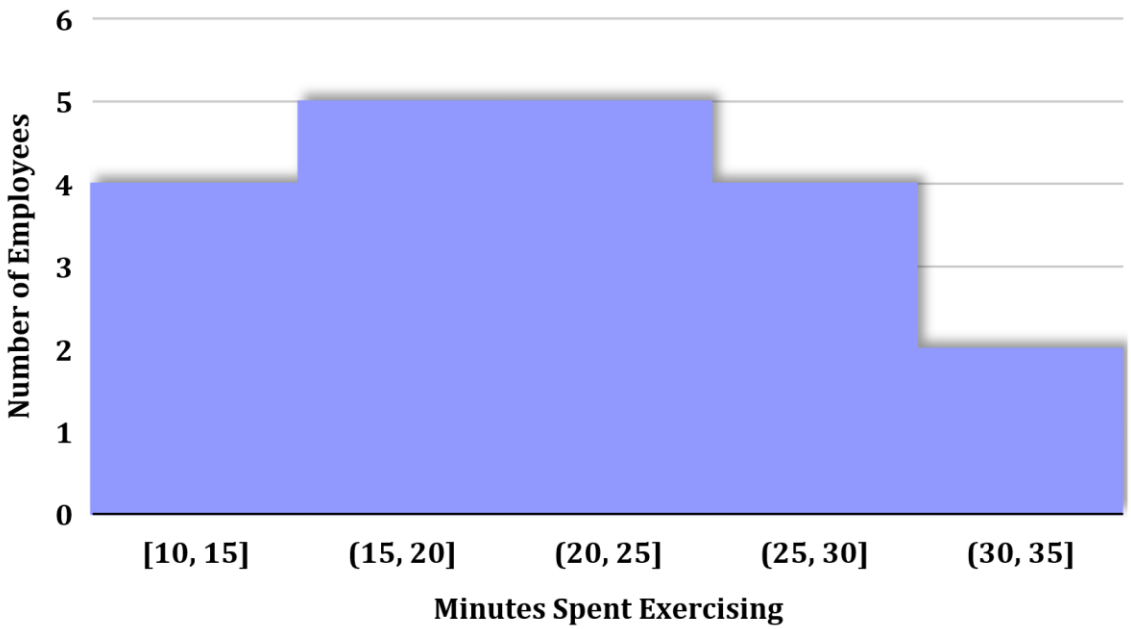
The range is $35 - 10 = 25$. We'll divide the data set into 5 classes, which means the class width will be

$$\frac{25}{5} = 5$$

Since 10 is the minimum value, we can use it as a starting point, or the lower limit of the first class. So our table for the classes and their frequencies will be

Classes	Frequency
10 – 15	4
15 – 20	5
20 – 25	5
25 – 30	4
30 – 35	2

We don't need it for this particular problem, but we could also determine the midpoint of each class.

Classes	Frequency	Class Midpoint
10 – 15	4	12.5
15 – 20	5	17.5
20 – 25	5	22.5
25 – 30	4	27.5
30 – 35	2	32.5

Now we can use the table to draw the histogram.